

Karthik C R

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PROFESSIONAL SUMMARY

Software engineering student with a strong foundation in systems programming, algorithms, and backend development, built through coursework in operating systems, networks, and software architecture, and validated by two published research papers. Proficient in Java, Python, and C++, with hands-on experience across the full delivery cycle from design, testing, optimization, and deployment in small, cross-functional teams. Quick to learn unfamiliar systems, comfortable with modern AI-assisted development workflows, and driven to build clean, efficient, production-ready software.

EDUCATION

Vellore Institute of Technology, Chennai Expected June 2027
Integrated M.Tech in Software Engineering CGPA: 8.86
Relevant Coursework: Operating Systems, Computer Networks, Object-Oriented Programming, Software Architecture and Design, Software Engineering, Software Testing and Quality Assurance, Database Management Systems (DBMS), and Data Structures and Algorithms (DSA).

SKILLS

Languages	Java, Python, SQL
Backend	Spring Boot, REST APIs, OOP, Multithreading, JUnit
Systems	Operating Systems, Memory Management, TCP/IP
Mobile / Web	Flutter, React, HTML, CSS, JavaScript
Gen AI / LLM	Prompt Engineering, RAG, Sentence-Transformer Embeddings, NLP (NER, Regex), Claude, Codex
Tools	Git, Apache JMeter, Selenium, Agile
Soft Skills	Problem Solving, Analytical Thinking, Team Collaboration, Clear Communication, and Debugging & Root Cause Analysis

EXPERIENCE

- Project Intern, Anna University Chennai** Aug 2025 – Sep 2025
- Built a fully offline inference pipeline with a 2-person team, enabling real-time crop disease detection at 97% accuracy in low-connectivity environments.
 - Cut inference latency and memory footprint by 30% through model optimization and system-level tuning, incorporating mentor feedback in weekly reviews.
- Deep Learning Research Intern, CCPS – VIT Chennai** Jun 2025 – Aug 2025
- Designed and evaluated dual-branch neural network pipelines with a 3-person research team, using structured experimentation and ablation testing to improve classification robustness.
 - Cut experimentation time by 18% by improving batching and pipeline efficiency, streamlining training workflows and resource use across experiments.

PROJECTS

- Automated B2B RFP Response System using Agentic AI and NLP** Jan 2026 – Apr 2026
- Built an end-to-end data pipeline with a 3-person team that cut proposal turnaround time 60–80% by automating RFP processing, PDF/OCR ingestion, spec extraction, and semantic SKU matching; owned agent design and core business logic using Claude and Codex for iteration and debugging.
 - Co-architected a multi-stage pipeline combining rule-based compliance validation with LLM-driven proposal generation, improving accuracy and consistency across enterprise tenders.
- Indoor Vehicle Tracking System (BLE-Based IoT Pipeline)** Jun 2025 – Nov 2025
- Built an ESP32-based BLE indoor localization system in C++ with signal filtering and RSSI outlier suppression, cutting mean positioning error by 40% under multipath interference.
 - Engineered a low-latency edge-to-cloud telemetry and data pipeline over Wi-Fi/HTTP, achieving 200 ms processing latency and 99%+ reliable data transmission on resource-constrained hardware.
- Scalable Multithreaded TCP Server (Java)** Aug 2025 – Sep 2025
- Engineered a concurrent TCP server using `ThreadPoolExecutor` with bounded queue and backpressure control (`CallerRunsPolicy`) to manage high-volume client connections.
 - Load-tested with up to 10,000 concurrent virtual users via Apache JMeter, analyzing throughput, latency, and thread pool saturation under peak traffic.

PUBLICATIONS

- Fusing Appearance and Vein Morphology Using Dual-Branch Deep Networks for Accurate Medicinal Plant Identification.** *Frontiers in Artificial Intelligence*, Vol. 9, April 2026. [\[DOI\]](#)
- Hyperparameter Tuning Strategies for Robust Machine Learning in Android Malware Detection.** *Security and Privacy*, Wiley, 2025. [\[DOI\]](#)